

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

M. Arshini 19040432 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Arshini

$25 + 25 + 25 + 12 + 8$
2 (95)
10

Annexure A

Constraint-Based Problem Solving

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1. A hospital needs to assign 3 nurses (N1, N2, N3) to 3 wards (ICU, General, Emergency) such that:

Constraints:

- i. N1 cannot be assigned to Emergency ward
- ii. N2 must be assigned before N3 (ICU < General < Emergency)
- iii. Only one nurse per ward
- iv. N3 cannot be assigned to ICU
- v. All nurses must be assigned exactly one ward

Apply Backtracking Search to find a valid assignment, showing all intermediate steps and constraint checks. State the final valid schedule.

2. A cleaning robot operates in a 5x5 room-grid from Start (S) to a docking Goal (G), with a limited battery allowing at most 8 moves.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1 (battery unit)
- iii. Cannot pass through furniture (obstacle) cells
- iv. The robot must reach G within the battery limit

Using the above constraints and Depth-Limited Search (with the battery limit as depth bound), show step-by-step node expansion and determine whether a valid path exists within the limit, along with its cost.

Hospital Nurse Assignment using Backtracking search

variables

* Nurses: N_1, N_2, N_3

Wards

* ICU
* General
* Emergency

constraints:-

* $N_1 \neq \text{Emergency}$

N_2 must be assigned before N_3

Order: ICU < General < Emergency

One nurse per ward

Every nurse gets exactly one ward.

Step 1:- Assign N_1

Try ICU

$N_1 \rightarrow \text{ICU} \checkmark$

constraint check

~~$N_1 \neq \text{Emergency}$~~

~~ward unique~~

current assignment

N_1

N_2

N_3

ward

ICU

—

—

Step 2:- Assign N2

Available wards:

General

Emergency

Try General

N2 → General

Constraint check

Word unique

N2 before N3

Current Assignment

Nurse

N1

N2

N3

ward

ICU

General

Step 3:- Assign N3:-

Remaining ward

* Emergency

check constraints

N3 not ICU

ICU < General < Emergency

N2 = General

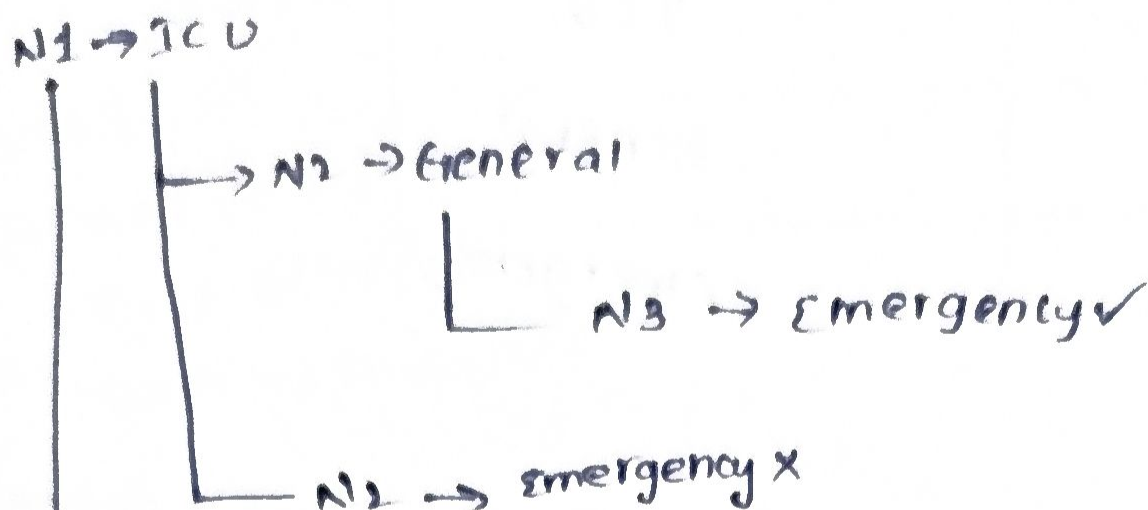
N3 = Emergency

General becomes before Emergency.

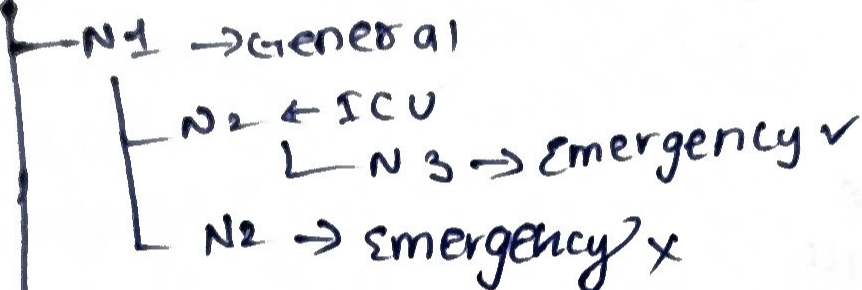
Assignment Accepted.

Search tree

start



N3 would have to take General, violating N2 before N3



N1 → Emergency X
(constraint violated)

Backtracking table

step	Assign	constraint check	result
1	N1 → ICU	N1 not emergency	✓
2	N2 → General	unique ward	✓
3	N3 → Emergency	N3 not ICU, N2 before N3	✓

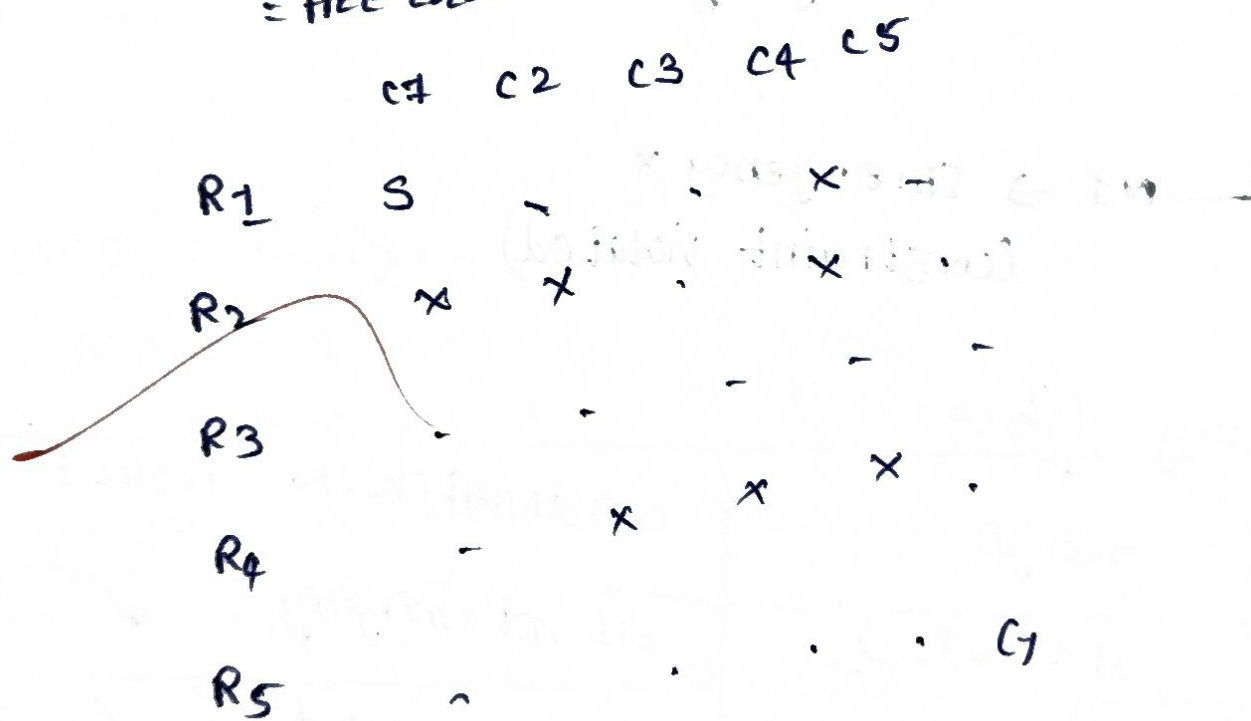
final valid schedule

nurse	ward
N1	ICU
N2	General
N3	Emergency

cleaning robot using Depth limited search (Depth limit = 8)

Assume the following 5x5 grid.

S = start
G = Goal
X = obstacle
= free cell



start = (1, 1)
Goal = (5, 5)
Depth limit = 8
move cost = 1

Step 1:-

current node:-

$(1, 1)$

Depth = 0

Expand

possible moves

Right $\rightarrow (1, 2)$ ✓

Down $\rightarrow$ obstacle x

visited

$(1, 1)$

Step 2:-

current node:-

$(1, 2)$

Depth = 1

Expand

Right $\rightarrow (1, 3)$ ✓

~~Left~~ $\rightarrow$ Already visited

~~Down~~ $\rightarrow$ obstacle x

visited

$(1, 1)$

$(1, 2)$

Step 3:- current node:-

$(1, 3)$

Depth = 2

Expand

down $\rightarrow (2,3)$
right $\rightarrow$ obstacle
left $\rightarrow$ visited

visited
(1,1)
(1,2)
(1,3)

step 4

current node:-
(2,3)
Depth = 2

Expand
down
right $\rightarrow (3,3)$ ✓
right
down $\rightarrow$ obstacle
visited
(2,4)
left $\rightarrow$ obstacle

visited

(1,1)
(1,2)
(1,3)
(2,3)

step 5:-

current node:-

(2,3)

Depth = 4

Expand

right $\rightarrow (3,4)$

left $\rightarrow (3,2)$

choose right

visited

(2,3)

step 6:-

current node:-

(3,4)

Depth = 5

Expand

Right $\rightarrow (3,5)$

down $\rightarrow$ obstacle

(3,4)

Step 7:-

current node

(3,5)

Depth = 6

Expand down $\rightarrow (4,5)$

Visited (3,5)

set 1:-

current node

(1,3)

depth = 2

current d

goal $\rightarrow$ (5,5) goal

depth = 5

goal found

depth limited search tree

L(1,1)

L(1,2)

L(1,3)

L(2,3)

L(3,3)

L(3,4)

L(3,5)

L(4,5)

L(5,5) goal

path found :- (1,1)

$\rightarrow$ (1,2)

$\rightarrow$ (1,3)

$\rightarrow$ (2,3)

$\rightarrow$ (3,3)

$\rightarrow$ (3,4)

$\rightarrow$ (3,5)

$\rightarrow$ (4,5)

$\rightarrow$ (5,5)

cost calculation

move	cost
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1

total cost = 8 battery unit



RUBRICS FOR CALCULATION:

Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

$25 + 25 + 25 + 12 + 8$
 95
 100